

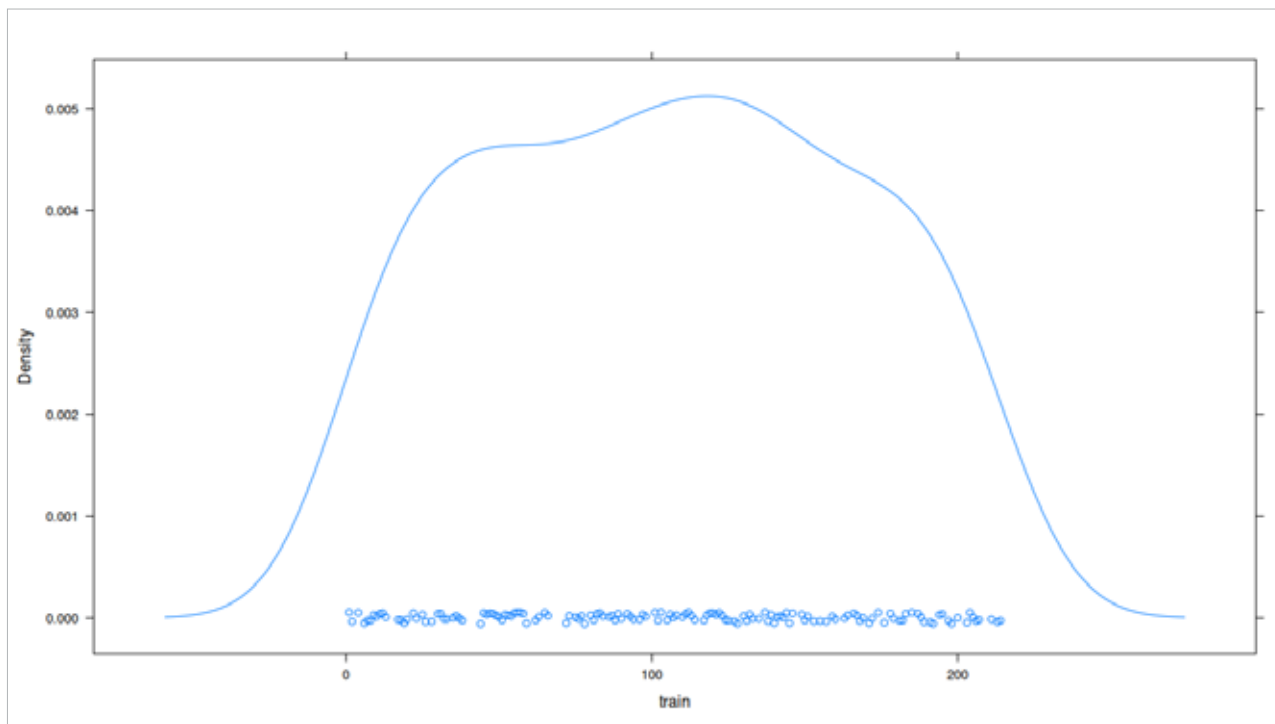


Figure 1: Density plot

```
> set.seed(501)
```

```
> fit <- bagFDA(Type ~ ., data)
```

No Information Rate : 0.3438

P-Value [Acc > NIR] : 2.194e-08

Kappa : 0.5757

predict()

The `predict()` function can classify observations with `mda`, and the data can be transformed into a matrix using the `confusionMatrix()` function as illustrated below:

```
> confusionMatrix(data = predict(fit, test[, -10]), reference = test[, 10])
```

Confusion Matrix and Statistics

	Reference						
Prediction	1	2	3	5	6	7	
1	20	6	6	0	0	0	
2	2	9	2	2	0	0	
3	0	0	0	0	0	0	
5	0	1	0	2	0	1	
6	0	0	0	0	3	0	
7	0	0	0	0	0	10	

Overall Statistics

Accuracy : 0.6875

95% CI : (0.5594, 0.7976)

McNemar's Test P-Value : NA

Statistics by Class:

	Class: 1	Class: 2	Class: 3	Class: 5	Class: 6	Class: 7
Sensitivity	0.9091	0.5625	0.000	0.50000	1.00000	0.9091
Specificity	0.7143	0.8750	1.000	0.96667	1.00000	1.0000
Pos Pred Value	0.6250	0.6000	NaN	0.50000	1.00000	1.0000
Neg Pred Value	0.9375	0.8571	0.875	0.96667	1.00000	0.9815
Prevalence	0.3438	0.2500	0.125	0.06250	0.04688	0.1719
Detection Rate	0.3125	0.1406	0.000	0.03125	0.04688	0.1562
Detection Prevalence	0.5000	0.2344	0.000	0.06250	0.04688	0.1562
Balanced Accuracy	0.8117	0.7188	0.500	0.73333	1.00000	0.9545

skim()

The `skimr` package provides summary statistics for a data set based on the principles of least surprise. You can install and load the library package as follows: